1. Background and Objective

Need to develop a comprehensive, browser-based IT Department Management System to streamline, centralize, and automate the management of IT operations, inventory, procurement, contracts, invoices, maintenance activities, and budget utilization.

The software will be designed exclusively for internal use by authorized IT department personnel within our organization. It will operate on a dedicated in-house server and will be accessible through web browsers on the organization's internal network.

2. Core Functional Requirements

The system should consist of the following interconnected modules:

1. **Employee Management:** Maintain employee records, departmental information, contact details, and relevant IT-related assignments.
2. **IT Inventory and Asset Management:** Maintain a centralized inventory database containing asset descriptions, categories, quantities, serial numbers, locations, and other relevant information. Each asset will be assigned a unique asset code manually by me to ensure proper identification and tracking.
3. **Purchase Order and Contract Management:** Maintain complete records of purchase orders, contracts, vendors, financial details, supporting documents, commencement dates, expiry dates, and renewal information. The system should generate expiry notifications and reminders before contracts or purchase orders reach their expiry dates.
4. **Invoice Management:** Record and manage invoices linked to their respective purchase orders, contracts, vendors, and budget allocations. The system should maintain payment status, invoice amounts, dates, and outstanding balances.
5. **Software and Maintenance Management:** Maintain records of software licenses, subscriptions, renewals, IT equipment maintenance, repair activities, service agreements, and associated expenses.
6. **Inventory Issuance and Return Management:** Record the issuance and return of IT assets to employees or departments, maintain complete transaction histories, and track the current status and location of each asset.
7. **Annual Budget Management:** Maintain annual IT budgets, allocated amounts, committed expenditures, actual expenditures, utilized budgets, and remaining balances. The system should provide budget utilization reports by financial year, purchase order, contract, vendor, and relevant expense category.

3. Integrated Database and Global Search

All modules must be interconnected through a centralized relational database. Information should not be maintained in isolated or duplicate records.

The system must provide a powerful global search facility. For example, when I enter a purchase order number, the system should retrieve all related information on a single screen, including:

- Purchase order details and associated contract information.
- Vendor details and relevant supporting documents.
- Related invoices, payment records, and financial information.
- Items purchased and associated inventory asset codes.
- Inventory issuance and return records, where applicable.
- Maintenance or software-related records linked to the purchase.
- Contract or purchase order expiry dates.
- Relevant budget allocation, utilization, and remaining balance.

The objective is to make all relevant information accessible with a single search, minimizing manual record retrieval and improving operational efficiency.

4. Document Generation and Downloadable Forms

The system should support standardized document generation using predefined organizational formats.

The required documents include:

- Purchase order details and reports.
- Contract details and reports.
- Inventory issuance forms.
- Inventory return forms.
- Invoice and expenditure reports.
- Budget utilization reports.
- Asset registers and inventory reports.

The generated documents should follow the specific formats used by our organization and be downloadable in appropriate formats, such as PDF and Excel, wherever applicable.

5. Asset Coding and Transaction Tracking

The system must allow me to manually assign unique asset codes to IT inventory items. These codes should be used consistently across inventory records, issuance forms, return forms, maintenance records, and other relevant transactions.

The system should maintain a complete history of each asset, including its procurement source, purchase order, acquisition cost, assigned employee, issuance date, return date, maintenance history, and current status.

6. Security and Access Control

As this software is intended exclusively for internal IT department use, it should include secure user authentication, role-based access permissions, and appropriate restrictions on sensitive financial and procurement information.

The system should also maintain audit logs of important activities, including record creation, modification, deletion, inventory issuance, returns, and financial updates. Regular database backups and recovery procedures must be implemented to protect organizational data.

7. Technical and Deployment Requirements

The proposed application should be:

- Browser-based and accessible through the internal organizational network.
- Hosted on a dedicated in-house computer or server.
- Supported by a centralized relational database.
- Modular, scalable, and easy to maintain.
- Designed with a user-friendly interface and dashboard.
- Capable of generating reports, downloading documents, and exporting data.
- Protected through authentication, authorization, audit logging, and regular backups.

8. Requested Development Approach

I would appreciate step-by-step technical guidance for planning, designing, developing, testing, deploying, and maintaining this software.

The development process should begin with requirements analysis and database schema design, followed by selection of an appropriate technology stack, interface design, module development, integration, reporting, security implementation, and deployment on the dedicated in-house server.

I would prefer a practical, structured approach in which each development phase is explained clearly, with implementation instructions, database tables and relationships, example code where required, and testing procedures.